PS Response: Problem Set 1

Applied Stats/Quant Methods 1

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Question 1: Education

A school counselor was curious about the average of IQ of the students in her school and took a random sample of 25 students' IQ scores. The following is the data set:

```
y <- c(105, 69, 86, 100, 82, 111, 104, 110, 87, 108, 87, 90, 94, 113, 112, 98,
    80, 97, 95, 111, 114, 89, 95, 126, 98)
```

1. Find a 90% confidence interval for the average student IQ in the school.

```
mean_y <- mean(y) # sample mean
n <- length(y)    # sample size == 25
s <- sd(y)        # sample standard deviation
SE <- s/sqrt(n)   # standard error of the mean
alpha <- 0.10
df<- n-1          # degrees of freedom == 24

# 90% two-sided CI: t_{1 - alpha/2, df}
tcrit_1 <- qt(1 - alpha/2, df) #two-sided 90% CI -> qt(0.95.24)
tcrit_1           # == 1.710882

ci_90_lower <- mean_y - tcrit_1 * SE
ci_90_upper <- mean_y + tcrit_1 * SE
```

```
> print(round(mean_y,2))
[1] 98.44
> print(round(n,2))
[1] 25
```

```
> print(round(s,2))
[1] 13.09
> print(round(SE, 2))
[1] 2.62
> print(round(tcrit_1,2))
[1] 1.71
> print(round(ci_90_lower,2))
[1] 93.96
> print(round(ci_90_upper,2))
[1] 102.92
```

```
cat("90% CI = (",
    round(ci_90_lower, 2), ",", round(ci_90_upper, 2), ")\n")
```

The sample mean was 98.44 with $n = 25$, $s = 13.09$, and standard error 2.62. Using a t-distribution with $df = 24$ and $t_{\text{crit, } 0.95,24} = 1.71$, the 90% confidence interval for the average IQ in the school is:

$$(93.96,\ 102.92).$$

2. Next, the school counselor was curious whether the average student IQ in her school is higher than the average IQ score (100) among all the schools in the country.

   Using the same sample, conduct the appropriate hypothesis test with $\alpha = 0.05$.

```
mu0 <- 100          # H0: mu=100, H1: mu > 100
t_stat <- (mean_y - mu0) / SE
t_crit <- qt(0.95, df) # because alpha is 0.05
p_value <- 1 - pt(t_stat,df)
```

```
> print(round(n, 2))
[1] 25
> print(round(mean_y, 2))
[1] 98.44
> print(round(s, 2))
[1] 13.09
> print(round(SE, 2))
[1] 2.62
> print(round(t_stat, 2))
[1] -0.6
> print(round(t_crit, 2))
[1] 1.71
> print(round(p_value, 2))
[1] 0.72
```

I conducted a one-sided t-test with $H_0 : \mu = 100$ and $H_1 : \mu > 100$. The sample mean was 98.44 with $n = 25$, $s = 13.09$, and standard error 2.62. The test statistic was $t = -0.596$, the critical value at $\alpha = 0.05$ was 1.711, and the one-sided p-value was 0.72. Since $p > 0.05$, we fail to reject the null hypothesis and conclude that there is no statistical evidence that the school mean IQ is greater than 100.

Or, I can simply use ***t.test*** function to get an answer.

```r
print(round(t_stat, 2))
```

```
One Sample t-test

data:  y
t = -0.59574, df = 24, p-value = 0.7215
alternative hypothesis: true mean is greater than 100
95 percent confidence interval:
93.95993      Inf
sample estimates:
mean of x
98.44
```

Question 2: Political Economy

Researchers are curious about what affects the amount of money communities spend on addressing homelessness. The following variables constitute our data set about social welfare expenditures in the USA.

`State`	*50 states in US*
`Y`	*per capita expenditure on shelters/housing assistance in state*
`X1`	*per capita personal income in state*
`X2`	*Number of residents per 100,000 that are "financially insecure" in state*
`X3`	*Number of people per thousand residing in urban areas in state*
`Region`	*1=Northeast, 2= North Central, 3= South, 4=West*

Explore the `expenditure` data set and import data into `R`.

```r
expenditure <- read.table("https://raw.githubusercontent.com/ASDS-TCD/StatsI_2025/main/datasets/expenditure.txt", header=T)
expenditure
```

- Please plot the relationships among *Y*, *X1*, *X2*, and *X3*? What are the correlations among them (you just need to describe the graph and the relationships among them)?

```r

#Select the relevant variables
vars <- expenditure[, c("Y", "X1", "X2", "X3")]

#(1) Pairwise scatterplot matrix
pdf("q2_pairs.pdf", width = 7, height = 7)
pairs(vars, main = "Pairwise Plots: Y, X1, X2, X3")
dev.off()

#(2) Scatterplots: Y vs each X
pdf("q2_Y_vs_Xs.pdf", width = 9, height = 3)
op <- par(mfrow = c(1,3))
plot(vars$X1, vars$Y, xlab="X1", ylab="Y", main="Y vs X1", pch=19, col="
    gray40")
plot(vars$X2, vars$Y, xlab="X2", ylab="Y", main="Y vs X2", pch=19, col="
    gray40")
plot(vars$X3, vars$Y, xlab="X3", ylab="Y", main="Y vs X3", pch=19, col="
    gray40")
par(op)
dev.off()

#(3) Correlation matrix
cor_mat <- cor(vars, use = "pairwise.complete.obs", method = "pearson")
round(cor_mat, 3)

cat("corr(Y, X1) =", round(cor_mat["Y","X1"], 3), "\n")
cat("corr(Y, X2) =", round(cor_mat["Y","X2"], 3), "\n")
```

```
> round(cor_mat, 3)
Y    X1    X2    X3
Y  1.000 0.532 0.448 0.464
X1 0.532 1.000 0.206 0.595
X2 0.448 0.206 1.000 0.221
X3 0.464 0.595 0.221 1.000

> cat("corr(Y, X1) =", round(cor_mat["Y","X1"], 3), "\n")
corr(Y, X1) = 0.532
> cat("corr(Y, X2) =", round(cor_mat["Y","X2"], 3), "\n")
corr(Y, X2) = 0.448
> cat("corr(Y, X3) =", round(cor_mat["Y","X3"], 3), "\n")
corr(Y, X3) = 0.464
```

Figure 1: Pairwise scatterplots among Y, X_1, X_2, and X_3.

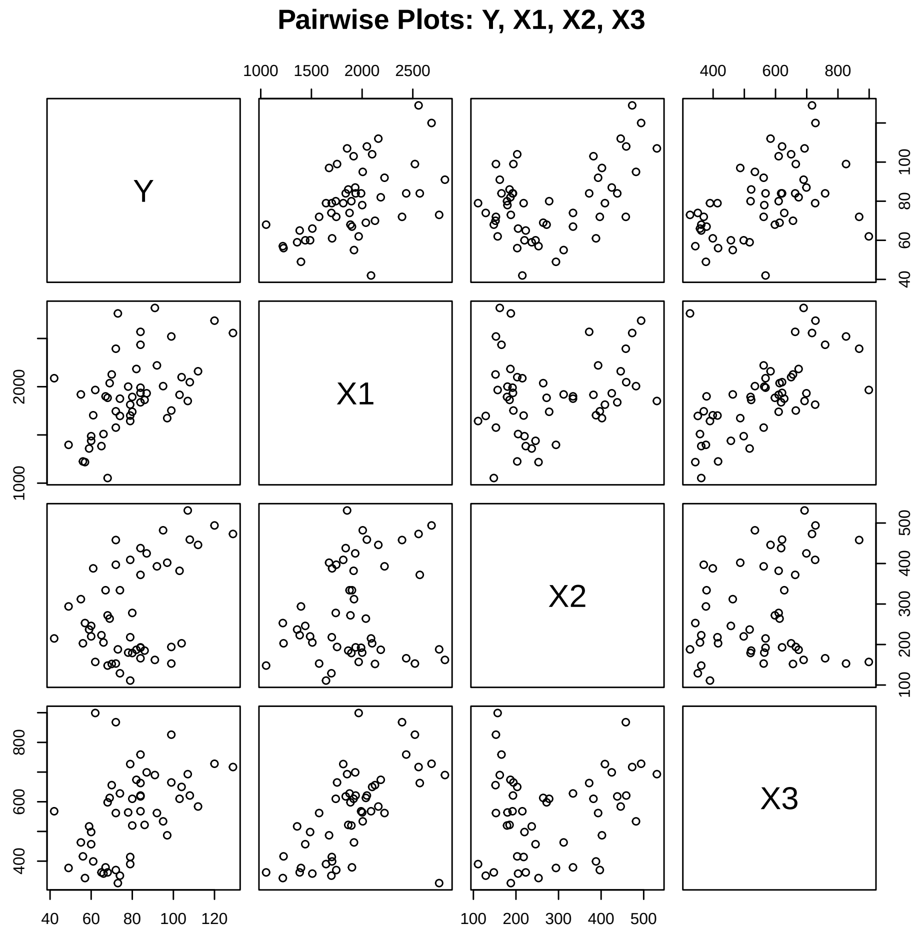
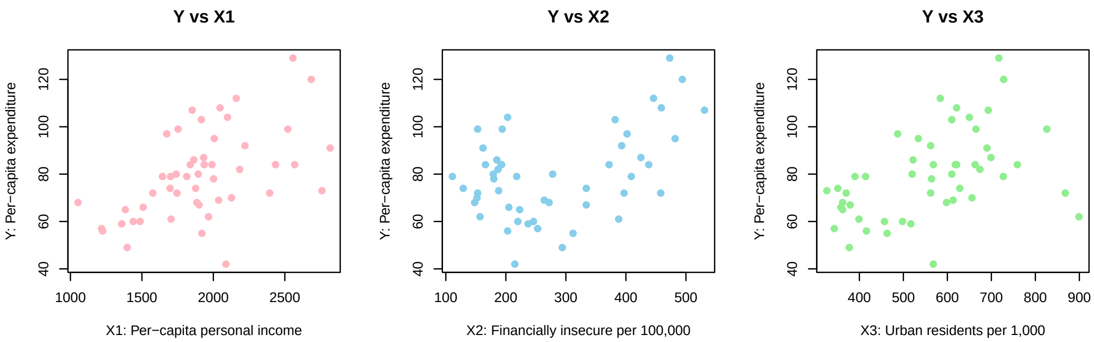


Figure 2: Scatterplots of Y vs X_1, X_2, and X_3.



From the scatterplots and the Pearson correlation matrix, all relationships between Y and X_1, X_2, and X_3 are positive. The association between Y and X_1 ($r = 0.532$) is the strongest, indicating that states with higher per-capita income tend to spend more on housing assistance. Meanwhile, X_2 (financial insecurity) and X_3 (urbanization) show moderate positive relationships with Y ($r = 0.448$ and $r = 0.464$ respectively). Overall, higher income and greater urbanization appear related to higher per-capita expenditure on housing assistance.

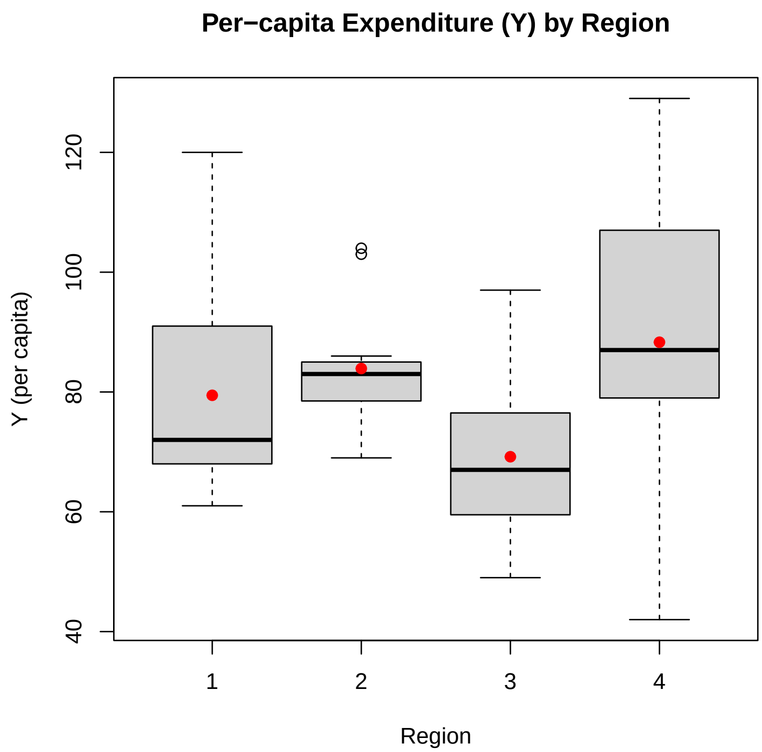
- Please plot the relationship between Y and *Region*? On average, which region has the highest per capita expenditure on housing assistance?

```
expenditure$Region <- factor(expenditure$Region)
# Boxplot
pdf("q2_Y_by_region.pdf", width=6, height=6)
boxplot(Y ~ Region, data=expenditure,
        main="Per-capita Expenditure (Y) by Region",
        xlab="Region", ylab="Y (per capita)")
region_means <- tapply(expenditure$Y, expenditure$Region, mean, na.rm =
    TRUE)
points(1:length(region_means), region_means, col="red", pch=19)
dev.off()

# Print region means
print(round(region_means,2))
```

```
> print(round(region_means,2))
    1     2     3     4 
79.44 83.92 69.19 88.31 
```

Figure 3: Boxplot of Y by Region. Red points indicate regional means.



I chose a **boxplot** because it clearly shows the distribution of Y within each region, including the spread, median, and potential outliers, and allows for easy comparison across groups. The red dots represent the regional means, making it straightforward to identify which region spends the most. From the output, **Region 4 (West)** has the highest mean per-capita expenditure on housing assistance, while **Region 3 (South)** shows the lowest. This suggests that western states tend to allocate more resources to housing assistance compared to southern states.

- Please plot the relationship between Y and *X1*? Describe this graph and the relationship. Reproduce the above graph including one more variable *Region* and display different regions with different types of symbols and colors.

```
pdf("q2_Y_vs_X1.pdf", width=6, height=6)
plot(expenditure$X1, expenditure$Y,
     main="Y vs X1",
     xlab="Per-capita personal income (X1)",
     ylab="Per-capita expenditure (Y)",
     pch=19, col="gray40")
abline(lm(Y ~ X1, data=expenditure), lty=2, lwd=2)
```

```r
dev.off()

# Scatterplot Y vs X1 by Region
pdf("q2_Y_vs_X1_by_region.pdf", width=6.5, height=6)
cols <- c("blue", "orange", "green", "purple")
pchv <- c(16, 17, 15, 18)
plot(expenditure$X1, expenditure$Y,
     main="Y vs X1 by Region",
     xlab="Per-capita personal income (X1)",
     ylab="Per-capita expenditure (Y)",
     col=cols[as.integer(expenditure$Region)],
     pch=pchv[as.integer(expenditure$Region)])
abline(lm(Y ~ X1, data=expenditure), lty=2, lwd=2)
legend("bottomright",
       legend=levels(expenditure$Region),
       col=cols, pch=pchv, bty="n",
       title="Region")
dev.off()
```

Figure 4: Scatterplot of Y vs X_1 with fitted regression line.

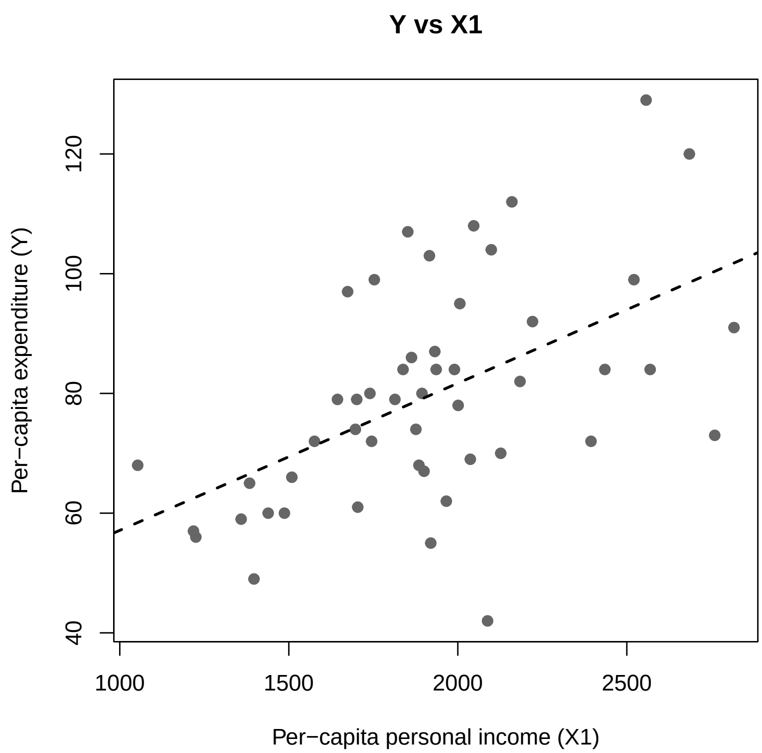
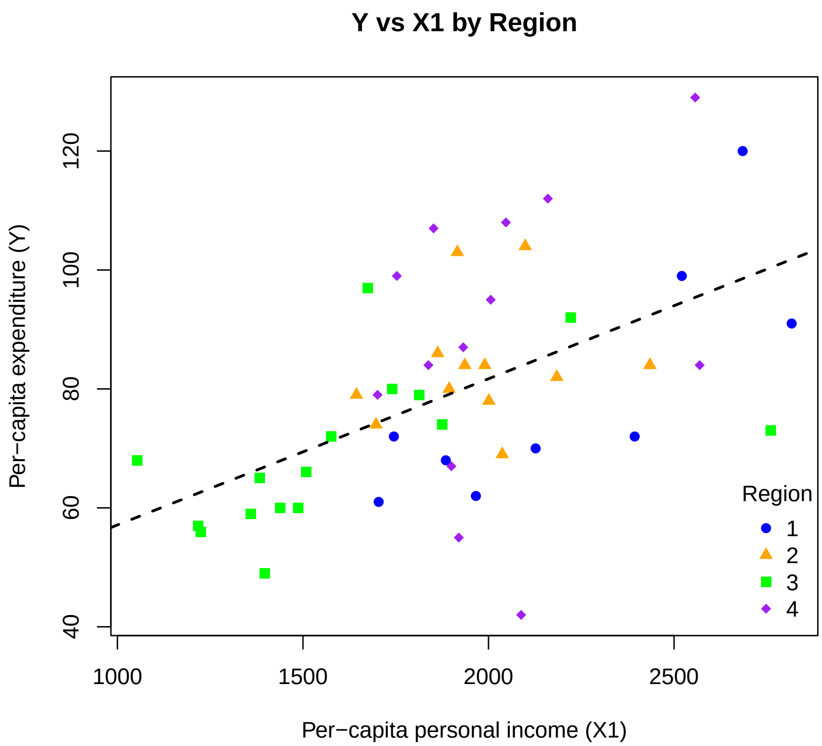


Figure 5: Scatterplot of Y vs X_1 by Region. Colors and symbols distinguish regions.



The scatterplot of Y versus X_1 shows a moderate positive relationship: states with higher per-capita income tend to spend more on housing assistance. When we add *Region*, the points are distinguished by color and symbol, revealing that while the overall trend is positive, regional differences are also visible. In particular, **Region 4 (West)** and **Region 1 (Northeast)** generally cluster toward higher income and expenditure levels, whereas **Region 3 (South)** tends to have lower values for both.